

Answer Key — The Microscope and the Discovery of the Cell

Final answer with a one-line reason. Full methods are in the Notes' worked examples.

Objective

- Q1 **(b) Robert Hooke** — Hooke named cells after seeing cork under his microscope (1665).
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- Q2 **(a) Father of Microbiology** — he first clearly saw and described living microbes.
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- Q3 **(b) lentil seed** — a lens is shaped like a lentil — thick in the middle, thin at the edges.
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- Q4 **(c) electron microscope** — the electron microscope reaches about a million times.
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- Q5 **(b) False** — they were dead, empty box-like compartments.
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- Q6 **micrographia** — Hooke's 1665 book Micrographia.
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- Q7 **magnifying** — the curved water surface magnifies the letters.
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Short answer

- Q8 The eye can only see objects above a certain size, so microbes were invisible. The invention of the microscope (better lenses) let us see them.
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- Q9 Light microscope $\approx$ 100–400 times; electron microscope $\approx$ 10,00,000 (a million) times.
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- Q10 Many small, empty, honeycomb-like compartments; he named each one a "cell".
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Long / application

- Q11 Hooke (1665) built a microscope, observed cork, named the "cell", and published Micrographia. Leeuwenhoek (1660s) made stronger lenses and was first to see living microbes (bacteria, blood cells) — the Father of Microbiology. Together they opened the hidden living world.
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- Q12 **Advantage:** low cost makes microscopy accessible to many people/students. **Limitation:** lower magnification/detail than high-powered lab microscopes. Still valuable because it lets more people observe and learn about the microscopic world.
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